
FastPartz

Security & SD-WAN Solution

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Scope: 4 branches (UAE) + offsite backup (optional)

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1. Executive Summary

The Problem

FastPartz operates 4 branches across the UAE, each running physical Windows Servers with vendor-supplied accounting software backed by MS SQL Server. The current network posture is **critically vulnerable**:

- **Public IP addresses** assigned directly to servers
- **RDP (TCP 3389)** open to the entire internet
- **SQL Server (TCP 1433)** open to the entire internet
- **No firewalls** — zero perimeter defense
- **No antivirus** — zero endpoint protection
- **No network segmentation** — flat network topology
- **No backup strategy** — no recovery capability
- **Direct server-to-server sync** over the public internet — unencrypted

Exposed RDP is the #1 initial access vector for ransomware gangs (Conti, LockBit, BlackCat, Akira). Exposed SQL ports enable direct data exfiltration and destruction. The absence of backups means any attack is a business-ending event.

What We Will Deliver

A **complete, layered security architecture** that:

- **Closes all public-facing ports** — zero attack surface from the internet
- **Encrypts all inter-branch traffic** — IPsec tunnels between all branches
- **Segments the network** — servers, workstations, and guests are isolated from each other
- **Hardens every endpoint** — antivirus, firewall, attack surface reduction on all servers
- **Implements 3-2-1 backups** — local, cross-branch, and offsite copies with immutability
- **Provides secure remote access** — VPN with multi-factor authentication
- **Monitors and alerts** — intrusion detection, DNS filtering, real-time alerts

Result: A posture that would withstand a targeted ransomware attack, built entirely on open-source software and commodity hardware.

2. Solution Overview

Layer	What Changes	Before	After
Perimeter	OPNsense firewall at each branch	No firewall, public ports open	Zero inbound ports, all traffic inspected
Transport	IPsec encrypted tunnels	Unencrypted sync over internet	AES-256 encrypted, certificate-authenticated
Segmentation	VLANs on managed switches	Flat network, everything reachable	Servers, workstations, guests isolated
Endpoint	Windows Defender + Enterprise AV	No AV, Defender disabled	Real-time protection, ransomware-specific rules
Backup	Local NAS + cross-branch + offsite	No backups at all	3 copies, 2 media types, 1 offsite, immutable
Access	IKEv2 VPN with MFA	Direct RDP over internet	Encrypted VPN, certificate + password + TOTP
Monitoring	IDS/IPS + DNS filtering + alerts	No visibility	Real-time threat detection, email/Telegram alerts

3. Software Stack

All core software is open-source or free-tier. No recurring license costs for infrastructure.

Software	Purpose	Key Features	Installed On
OPNsense	Firewall, router, VPN gateway, IDS/IPS, DNS filtering	Stateful firewall, IPsec/IKEv2, Suricata IDS/IPS, Unbound DNS with blocklists, DHCP, DynDNS client, web management GUI	N100 mini-PC (dedicated appliance per branch)
Suricata (OPNsense plugin)	Intrusion Detection & Prevention System (IDS/IPS)	ET Open + Abuse.ch rulesets (~35,000 rules), inline blocking, protocol anomaly detection	Runs within OPNsense on N100
Unbound DNS (OPNsense built-in)	DNS filtering & security	DNSBL blocklists (malware/phishing), DNSSEC validation, DNS-over-TLS upstream	Runs within OPNsense on N100
Windows Defender	Endpoint antivirus (built-in)	Real-time protection, Tamper Protection, Attack Surface Reduction (ASR) rules, cloud-delivered protection	Each Windows Server
ESET PROTECT or Bitdefender GravityZone	Enterprise antivirus (layered defense)	Central management console, ransomware shield, behavioral detection	Each Windows Server (~\$40/server/year)
Veeam Backup Community Edition	Server backup (free for up to 10 workloads)	Full/incremental backup, MS SQL VSS integration, application-aware, AES-256 encryption, email notifications	Master branch Windows Server
Borgbackup	Encrypted offsite backup	Deduplication, AES-256 encryption, compression, append-only mode (immutable), incremental	Offsite NAS (optional)
Restic	Cloud backup client	Backblaze B2 / Cloudflare R2 / AWS S3 integration, encryption, deduplication, retention policies	Master branch NAS or server
NXLog Community Edition	Windows log forwarding to syslog	Forwards Windows Security events (failed logins, account changes, service installs) to central syslog	Each Windows Server
Uptime Kuma	Availability monitoring dashboard	Ping/TCP/HTTP checks, beautiful web UI, push notifications, Docker-based	Master branch NAS (Docker) or any server

Enterprise AV Options (Recommended — Pick One)

Solution	Cost	Central Mgmt	Key Feature
ESET PROTECT	~\$40/server/year	Cloud console	Low resource usage, ransomware shield
Bitdefender GravityZone	~\$35/server/year	Cloud console	HyperDetect behavioral analysis

Total AV cost for 4 servers: ~\$140–160/year.

4. Hardware Bill of Materials

Per Branch — Required Items

Item	Spec	Purpose	Unit Cost (USD)
SD-WAN Edge (N100 Mini-PC)	Intel N100, 2× 2.5GbE NICs, 8 GB DDR5, 128 GB NVMe, fanless, AES-NI	Runs OPNsense — fire-wall, IDS/IPS, VPN, DNS filtering, DHCP. All branch traffic flows through this device.	\$150–200
Managed Switch	TP-Link TL-SG2428P or similar — 24× GbE RJ45, PoE+, VLAN (802.1Q), web management GUI	Network segmentation via VLANs. Trunk port to OPNsense. Access ports per VLAN.	\$200–300
Cat6 patch cables, misc	—	Wiring	\$30–50

N100 port usage: Port 1 = WAN (to broadband router), Port 2 = LAN (trunk to managed switch carrying all VLANs).

Managed switch requirements: RJ45 ports only (no SFP needed), VLAN support (802.1Q), PoE+ (for APs/cameras), web GUI for management. For smaller branches, TP-Link TL-SG2210P (8-port) is sufficient.

Active/Standby N100 for High Availability

For branches requiring zero downtime, deploy a second N100 as a cold standby:

Item	Purpose	Unit Cost
Standby N100 Mini-PC	Pre-configured with identical OPNsense config. If primary fails, swap and boot. Recovery in <5 minutes.	\$150–200

With OPNsense config backup (XML export), restoring to the standby takes minutes. The standby sits powered off until needed.

Optional Items

Item	Spec	Purpose	Unit Cost (USD)
Backup NAS (master branch only)	Synology DS223 (2-bay, RAID 1) + 2× WD Red Plus 4 TB	Local backup target for Veeam. Also serves as cross-branch replication target. Only needed at the master branch since all branches sync there.	\$400–500
Offsite NAS (Pakistan or other location)	Synology DS223 or TerraMaster F2-223 (2-bay) + 2× 4 TB drives	Receives encrypted backups from UAE over VPN tunnel. Provides geographic disaster recovery outside UAE.	\$400–500

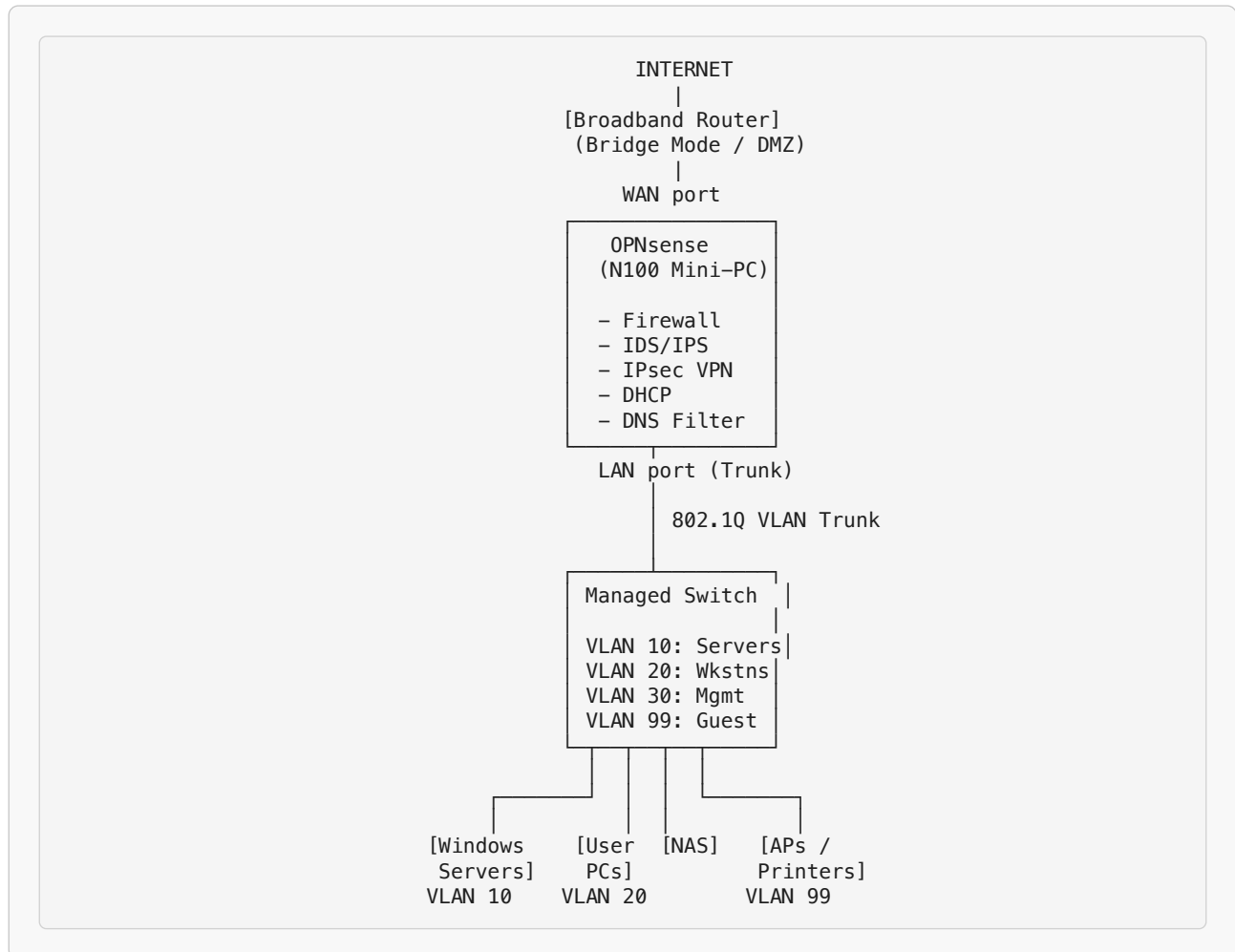
Cost Summary

Scenario	Per Branch	4 Branches	Optional	Grand Total
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Core (N100 + Switch)	\$380–550	\$1,520–2,200	—	\$1,520–2,200
Core + HA standby	\$530–750	\$2,120–3,000	—	\$2,120–3,000
Core + Backup NAS (master only)	—	\$1,520–2,200	\$400–500	\$1,920–2,700
Full (Core + HA + NAS + Offsite)	—	\$2,120–3,000	\$800–1,000	\$2,920–4,000

5. Network Architecture

Per-Branch Topology



Listing 1: Per-branch network topology

How Traffic Flows

OPNsense is the default gateway for every device. All traffic — internet, inter-branch, and inter-VLAN — passes through OPNsense for inspection and policy enforcement.

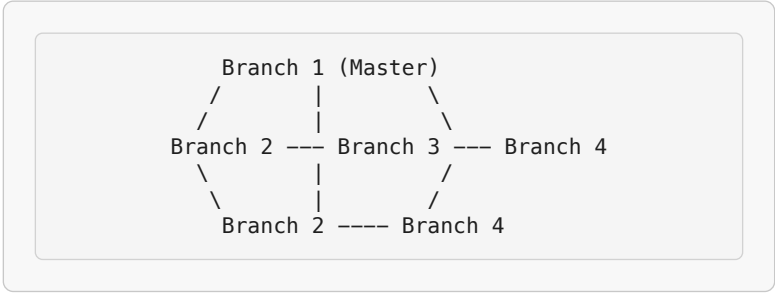
Traffic Type	Path	Protection Applied
Server-to-Server (inter-branch)	OPNsense → IPsec tunnel → remote OPNsense	AES-256 encryption + IDS/IPS
Server-to-Server (same branch)	Direct via switch (same VLAN)	Windows Firewall rules
Workstation to Internet	OPNsense → ISP	IDS/IPS + DNS blocklists
Workstation to Server (same branch)	OPNsense inter-VLAN routing	Firewall rules per port

Remote User to Server	IKEv2 VPN → OPNsense → Server	Encrypted + MFA authenticated
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N100 performance: Gigabit NAT ~940 Mbps, with Suricata IDS ~800 Mbps, IPsec AES-256-GCM ~700 Mbps. More than sufficient for branch office workloads.

Full-Mesh Inter-Branch Connectivity

All 4 branches are connected to each other via encrypted IPsec tunnels — **6 tunnels** total for full mesh:



Listing 2: Full-mesh IPsec tunnel topology (6 tunnels)

Every branch can reach every other branch directly. If one tunnel fails, other branches remain connected. Dead Peer Detection (DPD) automatically recovers tunnels within 60 seconds.

6. Security Features

What Gets Deployed at Each Branch

Feature	Description	Benefit
Stateful Firewall	OPNsense default-deny inbound. Only explicitly allowed traffic passes.	Closes all public ports (RDP, SQL, everything). Zero attack surface.
IDS/IPS	Suricata with ET Open + Abuse.ch rules (~35,000 signatures). Inline blocking mode.	Detects and blocks exploit attempts, malware downloads, C2 callbacks, port scans.
DNS Filtering	Unbound DNS with blocklists (malware, phishing, ads). DNSSEC validation. DNS-over-TLS upstream.	Blocks connections to known malicious domains before they happen.
Network Segmentation	VLANs isolate servers, workstations, management, and guests. Inter-VLAN firewall rules.	Limits blast radius. Compromised workstation cannot reach servers in other VLANs/branches.
Encrypted Transport	IKEv2 IPsec with AES-256-GCM, certificate authentication, Perfect Forward Secrecy.	All inter-branch data is encrypted. No eavesdropping or tampering possible.
Endpoint Protection	Windows Defender + ASR rules + enterprise AV (ESET/Bitdefender).	Multi-layer malware defense on every server.
Server Hardening	Disabled: SMBv1, NetBIOS, LLMNR, PowerShell v2. Enabled: Windows Firewall, audit logging, account lockout.	Reduces exploitable services to absolute minimum.
Secure Remote Access	IKEv2 VPN with certificate + password + TOTP (MFA). No client software needed.	Eliminates exposed RDP. Remote users must authenticate through VPN.
Backup with Immutability	3-2-1 backup: local + cross-branch + offsite. Immutable snapshots prevent ransomware encryption of backups.	Full recovery capability even in worst-case scenario.
Monitoring & Alerting	Availability checks, IDS alerts, backup status. Email + Telegram notifications.	Immediate awareness of security events or outages.

7. Network Segmentation (VLANs)

IP Addressing Scheme

Pattern: 10.<branch>.<vlan>.0/24

Branch	VLAN 10 (Servers)	VLAN 20 (Workstations)	VLAN 30 (Mgmt)	VLAN 99 (Guest/IoT)
Branch 1	10.1.10.0/24	10.1.20.0/24	10.1.30.0/24	10.1.99.0/24
Branch 2	10.2.10.0/24	10.2.20.0/24	10.2.30.0/24	10.2.99.0/24
Branch 3	10.3.10.0/24	10.3.20.0/24	10.3.30.0/24	10.3.99.0/24
Branch 4	10.4.10.0/24	10.4.20.0/24	10.4.30.0/24	10.4.99.0/24

VLAN Descriptions

VLAN	Name	What Connects Here	Access Rules
10	Servers	Windows Servers, NAS, SQL Server	Accessible from Workstations (SQL + file share only). Accessible from other branches via IPsec.
20	Workstations	User PCs, laptops	Can reach servers for SQL and file shares. Full internet access via OPNsense (inspected).
30	Management	OPNsense WebGUI, switch management	Only accessible by IT administrators. All other VLANs blocked from reaching this.
99	Guest / IoT	Guest WiFi, printers, cameras, IoT	Fully isolated. Internet access only (web). Cannot reach any internal VLAN.

Inter-VLAN Access Control

Default: DENY everything between VLANs. Only these specific flows are allowed:

From	To	Allowed Ports	Purpose
Workstations	Servers	TCP 1433 (SQL), TCP 445 (file shares)	Accounting software, file access
Workstations	Guest/IoT	TCP 9100, 631 (printing)	Printing
Management	All	All	IT administration
Guest/IoT	Internet only	TCP 80, 443	Web browsing for guests

Blocked: Guest/IoT can never reach Servers or Management. Workstations can never reach Management. No VLAN can access another except through explicit rules above.

8. Encrypted Inter-Branch Connectivity (IPsec SD-WAN)

How It Works

Each branch runs OPNsense which establishes encrypted IPsec tunnels to every other branch. The accounting software's SQL sync now travels through these encrypted tunnels instead of over the open internet.

Feature	Specification
Protocol	IKEv2 (modern, supports dynamic IPs, NAT traversal, mobile clients)
Authentication	X.509 Certificates from a private CA (no shared passwords)
Encryption	AES-256-GCM (hardware-accelerated on N100)
Key Exchange	ECDH Group 20 (384-bit, NIST P-384)
Perfect Forward Secrecy	Yes — new keys every hour, past sessions cannot be decrypted
Failover	Dead Peer Detection: detects failed tunnel in ~60 seconds, auto-recovers
NAT Traversal	Automatic (UDP 4500) — works behind any ISP NAT/router

Tunnel Matrix

#	Branch A	Branch B	Traffic
1	Branch 1	Branch 2	SQL sync, management
2	Branch 1	Branch 3	SQL sync, management
3	Branch 1	Branch 4	SQL sync, management
4	Branch 2	Branch 3	SQL sync, management
5	Branch 2	Branch 4	SQL sync, management
6	Branch 3	Branch 4	SQL sync, management

Certificate-Based Trust

A private Certificate Authority (CA) is created on the master branch OPNsense. Each branch receives a signed certificate. Only devices with certificates signed by this CA can establish tunnels. This is significantly stronger than pre-shared keys and cannot be brute-forced.

9. Dynamic DNS for Branch Discovery

Since UAE broadband uses dynamic public IPs, each branch OPNsense registers its current IP with a Dynamic DNS (DDNS) provider. IPsec peers find each other by hostname (FQDN) rather than IP address.

Feature	Detail
DDNS Provider	Dynu (recommended, free, 4 hostnames) or Cloudflare (if client owns a domain)
Branch FQDNs	<code>branch1.fastpartz.dynu.net</code> , <code>branch2.fastpartz.dynu.net</code> , etc.
Update Speed	DNS record updates within 60 seconds of IP change
Tunnel Recovery	If IP changes, tunnels recover automatically in 30–90 seconds via DPD
OPNsense Integration	Built-in DDNS client — no external scripts needed

Impact of ISP IP change: Brief tunnel interruption (~30–90 seconds). SQL sync retries automatically. No manual intervention needed.

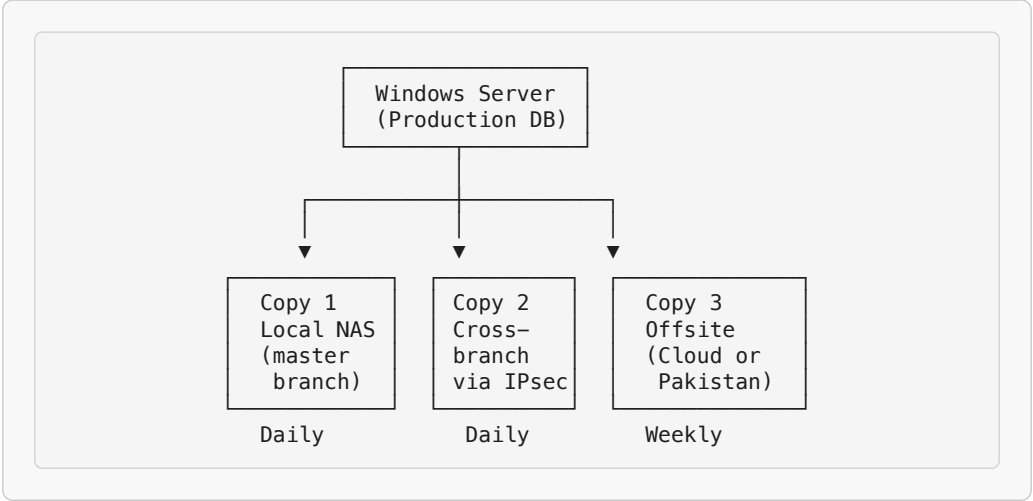
10. Windows Server Hardening

What Changes on Each Server

Category	Action	Why
Antivirus	Enable Windows Defender with Tamper Protection + ASR rules. Install enterprise AV (ESET/Bitdefender).	Multi-layer defense against malware and ransomware.
RDP	Disabled from internet. Accessible only via VPN + Management VLAN. NLA (Network Level Auth) required.	Eliminates the #1 ransomware entry point.
SQL Server	Bind to Server VLAN IP only (not 0.0.0.0).	SQL only reachable from internal network, not from any external source.
Windows Firewall	Enabled. Default-deny inbound. Allow only SQL (1433), SMB (445), RDP (from mgmt only).	Defense-in-depth even if network firewall is bypassed.
Disabled Protocols	SMBv1 (WannaCry vector), NetBIOS, LLMNR, WPAD, PowerShell v2 (bypasses AMSI).	Removes unnecessary attack surface.
Account Security	Lockout after 5 failed attempts (30 min). 14-char minimum password. Separate admin accounts.	Prevents brute-force and limits damage from credential theft.
Application Control	AppLocker or WDAC: block executables from user profile, temp, and AppData folders.	Prevents ransomware from executing even if downloaded.
Patching	Automatic Windows Update for security patches. Monthly reboot schedule.	Closes known vulnerabilities promptly.
Audit Logging	Security events forwarded to central syslog via NXLog. Track logins, account changes, service installs.	Forensic trail and real-time alerting on suspicious activity.

11. Backup Strategy (3-2-1)

Three copies of data, on two different media types, with one copy offsite.



Listing 3: 3-2-1 backup architecture

Copy	Location	Method	Schedule	Immutability
1 — Local	Synology NAS at master branch	Veeam Community Edition (SQL-aware, free)	Daily incremental + weekly full	NAS snapshots (read-only, 7-day retention)
2 — Cross-branch	Partner branch NAS via IPsec tunnel	rsync/rcclone with encryption	Daily sync	NAS snapshots
3a — Cloud (recommended)	Backblaze B2 / Cloudflare R2 / AWS S3 Glacier	Restic with AES-256 encryption	Weekly full + daily incremental	Object Lock (30-day, cannot be deleted)
3b — Physical offsite (optional)	NAS at Pakistan location	Borgbackup with AES-256 encryption over VPN	Weekly full + daily incremental	Append-only mode (client cannot delete backups)

Cloud Offsite Options

Provider	Storage Cost	Egress Cost	Immutability	Best For
Backblaze B2	\$5/TB/month	\$0.01/GB	Object Lock	Best balance of cost and features
Cloudflare R2	\$15/TB/month	Free	No native lock	Frequent restores needed
AWS S3 Glacier Deep Archive	\$1/TB/month	\$0.09/GB	Vault Lock	Cheapest long-term archival

Backup Testing

Test	Frequency
Backup completion verification	Daily (automated alert on failure)

File-level restore	Monthly
Full server restore to test environment	Quarterly
Offsite restore	Semi-annually

12. Secure Remote Access

IKEv2 Roadwarrior VPN

Remote users connect to the branch OPNsense via IKEv2 VPN. **No client software needed** — Windows, macOS, iOS, and Android all have built-in IKEv2 VPN clients.

Feature	Detail
Protocol	IKEv2 (native on all modern OS)
Authentication	Certificate + Username/Password + TOTP (MFA)
Encryption	AES-256-GCM
Split Tunneling	Only branch traffic goes through VPN (internet browsing stays local)
VPN Subnet	10.10.10.0/24 (dedicated, isolated from all branch VLANs)
DNS	OPNsense Unbound (filtered — malware/phishing domains blocked)

Remote User Access Control

VPN users CAN access	VPN users CANNOT access
Servers (RDP, SQL, file shares)	Workstation VLAN
Only on authorized ports	Management VLAN
	Guest/IoT VLAN
	Other VPN users

Vendor/Support Access

Control	Detail
Time-limited accounts	VPN account with expiry date
Restricted access	Only specific server IPs and ports
Session logging	All VPN connections logged
Immediate revocation	Account disabled after support session

13. Ransomware Attack Surface Analysis & Mitigations

#	Attack Vector	Current Risk	Mitigation	Risk After
1	RDP Brute Force	CRITICAL	OPNsense blocks all inbound. RDP only via VPN + Management VLAN.	ELIMINATED
2	SQL Server Exploitation	CRITICAL	OPNsense blocks all inbound. SQL bound to VLAN 10 only.	ELIMINATED
3	Phishing / Email	HIGH	Email provider security + SPF/DKIM/DMARC + user awareness training.	MEDIUM
4	Malicious Downloads	HIGH	Suricata IDS/IPS + DNS blocklists + Windows Defender + Enterprise AV + AppLocker.	LOW
5	USB / Removable Media	MEDIUM	Group Policy: disable AutoRun. ASR rule blocks untrusted processes from USB.	LOW
6	Lateral Movement	CRITICAL	VLAN segmentation + inter-VLAN firewall + Windows Firewall + separate admin accounts.	LOW
7	Credential Theft	HIGH	MFA on VPN. Account lockout (5 attempts/30 min). Separate admin accounts. LSASS protection.	LOW
8	Supply Chain / Software	MEDIUM	Auto-patching. AppLocker/WDAC whitelist. Vendor access time-limited.	LOW
9	Insider Threat	MEDIUM	Least privilege. Audit logging. No shared accounts.	LOW
10	WiFi	MEDIUM	WPA3. Guest SSID on VLAN 99 (isolated from servers).	LOW
11	Printers / IoT	LOW-MEDIUM	VLAN 99 (isolated). Only print ports allowed from workstation VLAN.	LOW
12	DNS-Based Attacks	MEDIUM	DNSSEC validation. DNS-over-TLS. DNS blocklists on OPNsense.	LOW
13	Backup Destruction	CRITICAL	3-2-1 backup with immutable snapshots. Object Lock on cloud. Append-only offsite.	LOW
14	Unpatched Vulnerabilities	HIGH	Automated Windows Update. Monthly patch review.	LOW

Email / Phishing Protection

DNS-based email authentication will be configured on the company domain:

Record	Purpose
SPF	Specifies which mail servers can send email for your domain
DKIM	Cryptographically signs outgoing emails to prevent spoofing
DMARC	Tells receiving servers to reject emails that fail SPF/DKIM

Additional controls: block dangerous attachment types (`.exe` , `.bat` , `.ps1` , `.vbs` , `.js`) at the mail gateway.
Quarterly phishing awareness training for staff.

14. Monitoring & Alerting

What Gets Monitored

Check	What It Detects	Alert Method
IPsec tunnel status	Tunnel down between any two branches	Email + Telegram
IDS/IPS alerts	Exploit attempts, malware downloads, C2 callbacks	Email
Server availability	Windows Server offline or SQL unreachable	Email + Telegram
Backup job status	Backup failed or incomplete	Email + Telegram
Cross-branch sync	Replication failure	Telegram
Offsite backup	Cloud or Pakistan backup failure	Telegram
Disk space	NAS or server storage >80% full	Email
Failed logins	5+ failed login attempts on any server	Email

Alert Channels

Channel	Use Case	Why
Email	All alerts	Formal record, reaches everyone
Telegram Bot	Critical alerts (tunnel down, backup failed, server offline)	Instant mobile notification, free, works globally

Monitoring Dashboard

Uptime Kuma (free, open-source) provides a web dashboard showing the status of all branches, tunnels, servers, and backup jobs at a glance. Runs as a lightweight container on the master branch NAS or server.

15. Implementation Phases

Phase	Scope	Duration	Deliverable
1	Hardware procurement + OPNsense installation on all N100 devices	1 week	4 configured OPNsense appliances ready for deployment
2	Branch 1 (master) pilot — full VLAN deployment, firewall rules, IDS/IPS, DNS filtering	1 week	Branch 1 fully operational with segmented, protected network
3	Deploy remaining 3 branches + establish all 6 IPsec tunnels (full mesh)	1 week	All branches connected with encrypted tunnels. SQL sync working.
4	Windows Server hardening + AV deployment + backup setup on all servers	1 week	All servers hardened. Veeam backups running daily.
5	Remote user VPN + monitoring + offsite backup (cloud/Pakistan)	1 week	VPN with MFA working. Monitoring dashboard live. 3-2-1 backup complete.
6	Testing, verification, documentation, knowledge transfer	1 week	All tests passed. Runbooks delivered. Staff trained.

Total estimated deployment: 6 weeks

16. Open Questions

To be discussed and filled in during client meeting:

#	Question	Answer	Impact
1	What email provider do the branches use? (Gmail, local UAE provider, Microsoft 365, other?)		Determines email security controls (SPF/DKIM/DMARC setup, attachment filtering options)
2	Which branch is the “master” that others sync to? Or is it peer-to-peer sync?		Determines where NAS and Veeam are deployed, backup flow direction
3	How much data (GB/TB) is on each server?		Sizes NAS drives and cloud backup costs
4	How often do servers sync, and how much data changes daily?		Sizes IPsec tunnel bandwidth needs and backup window
5	What is the broadband speed at each branch (download/upload)?		Validates N100 is sufficient, sizes cross-branch replication window
6	Are broadband routers ISP-managed or client-owned? Can we set bridge mode?		Determines WAN configuration (bridge vs DMZ vs port forward)
7	Does the client own a domain name (e.g., fastpartz.ae)?		Determines DDNS strategy (Cloudflare with own domain vs Dynu free)
8	How many remote users need VPN access?		Sizes VPN configuration and user certificate management
9	Are there any printers, IP cameras, or IoT devices on the network?		Determines VLAN 99 scope and firewall rules
10	Is there WiFi at any branch? If so, what APs?		Determines VLAN tagging on APs and WiFi security config
11	Does the client have a Microsoft 365 or Google Workspace subscription?		Determines available email security features
12	What Windows Server version is currently running at each branch?		Determines patching strategy and feature availability (AppLocker, ASR)
13	Is the accounting software vendor supportive of network changes? Will they need access?		Determines vendor VPN access requirements
14	Is there a preference for cloud offsite (Backblaze/AWS) vs physical offsite (Pakistan NAS)?		Determines offsite backup architecture
15	What is the acceptable downtime window for maintenance/reboots?		Sets maintenance window for patching and updates

Appendix: Emergency Procedures

Suspected Ransomware Attack

1. **ISOLATE** — Disconnect affected server from network immediately (pull Ethernet cable)
2. **DO NOT reboot** — preserves forensic evidence in memory
3. **CHECK** — Are other branches affected? Are backups intact?
4. **RESTORE** — Recover from most recent clean backup (local NAS → cross-branch → offsite)
5. **INVESTIGATE** — Review Suricata logs and Windows Security logs for entry point
6. **HARDEN** — Address the vulnerability that was exploited

OPNsense Failure

1. Branch loses internet and inter-branch connectivity
2. **Bypass:** Set workstation gateway to broadband router directly (internet-only, no security)
3. **Restore:** Boot standby N100 (if available) or re-flash OPNsense + restore XML config backup
4. **Recovery time:** ~5 minutes with standby unit, ~30 minutes with fresh install

IPsec Tunnel Down

1. Check OPNsense dashboard → VPN → IPsec → Status Overview
2. Wait 2 minutes (DPD auto-recovery in progress)
3. If still down: check if remote branch has broadband outage
4. Try disconnect + reconnect from OPNsense GUI
5. Check if DynDNS has updated (remote IP may have changed)